Classical Mechanics - II

Assignment - 8

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Q (1.) Uniform string, $\mathcal{L} = \frac{1}{2}(\sigma u_t^2 - \tau u_x^2)$, $x \in [-l, l]$

Eqⁿ of motion:

$$\frac{\partial \mathcal{L}}{\partial u_t} = \sigma u_t \quad , \quad \frac{\partial \mathcal{L}}{\partial u_x} = -\tau u_x \quad , \quad \frac{\partial \mathcal{L}}{\partial u} = 0$$

Euler-Lagrange eqⁿ: $\partial_t(\sigma u_t) + \partial_x(-\tau u_x) = 0$

$$\Rightarrow \quad \sigma u_{tt} - \tau u_{xx} = 0 \iff \boxed{u_{tt} = c^2 u_{xx}}$$

⇒ Symmetries i.e., conserved quantities:

- $\mathcal{L}$ has no explicit t: time translation ⇒ Energy
- $\mathcal{L}$ has no explicit x: space translation ⇒ (pseudo)-momentum
- $\mathcal{L}$ depends only on ∂u, not on u: shift $u \to u + \epsilon$ ⇒ transverse momentum, current $j^\mu = (\sigma u_t, -\tau u_x)$

(#) Stress-Energy tensor:

Momenta: $\frac{\partial \mathcal{L}}{\partial u_t} = \sigma u_t$, $\frac{\partial \mathcal{L}}{\partial u_x} = -\tau u_x$, Then:

$$T^0_{\ 0} = \sigma u_t^2 - \mathcal{L} = \frac{1}{2}(\sigma u_t^2 + \tau u_x^2) \equiv \mathcal{H}$$

$$T^1_{\ 0} = (-\tau u_x) u_t = -\tau u_x u_t$$

$$T^0_{\ 1} = \sigma u_t u_x \quad , \quad T^1_{\ 1} = (-\tau u_x) u_x - \mathcal{L} = -\tau u_x^2 - \frac{1}{2}\sigma u_t^2 + \frac{1}{2}\tau u_x^2$$

$$= -\mathcal{H}$$

$$\Rightarrow T^{\mu}_{\ \nu} = \begin{pmatrix} \mathcal{H} & \sigma u_t u_x \\ -\tau u_x u_t & -\mathcal{H} \end{pmatrix}, \quad \mathcal{H} = \frac{1}{2}\sigma u_t^2 + \frac{1}{2}\tau u_x^2$$

Continuity Equations :

$$\partial_\mu T^{\mu}_{\ \nu} = -\partial \mathcal{L}/\partial x^\nu \big|_{\text{explicit}} = 0 \quad (\because \text{no explicit } t \text{ or } x)$$

$\Rightarrow \nu = 0$ (Energy) :

$$\therefore \boxed{\partial_t \mathcal{H} + \partial_x(-\tau u_x u_t) = 0} \longrightarrow (i)$$

$\Rightarrow \nu = 1$ (momentum) :

$$\boxed{\partial_t(\sigma u_t u_x) - \partial_x \mathcal{H} = 0} \longrightarrow (ii)$$

Define the momentum density $\mathcal{P} = -\sigma u_t u_x$, the eq^n: (ii) becomes $\Rightarrow \partial_t \mathcal{P} + \partial_x \mathcal{H} = 0$

$\Rightarrow$ Shift symmetry current: $\boxed{\partial_t(\sigma u_t) + \partial_x(-\tau u_x) = 0}$

(This is just the EOM)

Conserved quantities & Boundary conditions

Integrate each eq^n over $[-l, l]$:

$$E = \int_{-l}^{l} \frac{1}{2}(\sigma u_t^2 + \tau u_x^2)\, dx \Rightarrow \frac{dE}{dt} = [\tau u_x u_t]_{-l}^{l}$$

$$P = -\int_{-l}^{l} \sigma u_t u_x\, dx \Rightarrow \frac{dP}{dt} = -[\mathcal{H}]_{-l}^{l}$$

$$P_1 = \int_{-l}^{l} \sigma u_t\, dx \Rightarrow \frac{dP_1}{dt} = [\tau u_x]_{-l}^{l}$$

⇒ Dependence on boundary conditions:

- Energy: Fixed ends ($u_t = 0$) ⇒ conserved. Free ends ($u_x = 0$) ⇒ conserved. Periodic or infinite string ⇒ conserved. So, E is conserved for all the usual boundary conditions.

- Momentum P: conserved only if $\mathcal{H}(l) = \mathcal{H}(-l)$ (periodic) or fields vanish at the ends (infinite string). For fixed ends $\mathcal{H} = \frac{1}{2}\tau u_x^2 \neq 0$ at the ends & in general unequal, so P is not conserved i.e., the walls break translation invariance.

- Transverse momentum $P_\perp$: Conserved for free or periodic ends. For fixed ends the supports exert a transverse force τu_x, so it is not conserved.

⇒ Physical Interpretation:

- $\mathcal{H} = \frac{1}{2}\sigma u_t^2 + \frac{1}{2}\tau u_x^2$ ⇒ Kinetic + Elastic pot. energy per unit length

- $-\tau u_x u_t$ ⇒ Energy flux (Power). The force exerted on the right part by the left part is $-\tau u_x$, and the power is force × velocity $= -\tau u_x u_t$.

- $\mathcal{P} = -\sigma u_t u_x$ is the wave (pseudo-) momentum density, not mechanical momentum (the string moves only transversely). For a right-moving wave $u = f(x - ct)$: $\mathcal{P} = \sigma c (f')^2 > 0$, $\mathcal{H} = \tau (f')^2$, so $p = E/c$, like a photon.

- The momentum flux is $\mathcal{H}$ ⇒ Energy density acts like a pressure that carries momentum along the string.

- σ_{xt} is the transverse momentum density, & $-\tau_{xx}$ is its flux (minus the transverse force).

(2.) To show $\partial_\mu T^\mu_\nu = -\dfrac{\partial \mathcal{L}}{\partial x^\nu}$ for $\mathcal{L}(\phi, \partial_t\phi, \partial_i\phi, t, x^i)$

Differentiate $T^\mu_\nu = \dfrac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\nu\phi - \delta^\mu_\nu \mathcal{L}$ with the product rule:

$$\partial_\mu T^\mu_\nu = \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right)\partial_\nu\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu\partial_\nu\phi - \partial_\nu \mathcal{L} \,,$$

(here $\partial_\nu \mathcal{L}$ is the Total derivative of $\mathcal{L}$ along x^ν, coming from $\partial_\mu(\delta^\mu_\nu \mathcal{L})$)

⇒ Euler-Lagrange: $\partial_\mu \dfrac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} = \dfrac{\partial \mathcal{L}}{\partial\phi}$

Substitute into the 1st term:

$$\partial_\mu T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial\phi}\partial_\nu\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu\partial_\nu\phi - \partial_\nu \mathcal{L} \quad \rightarrow (i.)$$

Chain rule for the total derivative of $\mathcal{L}$:

$$\partial_\nu \mathcal{L} = \frac{\partial \mathcal{L}}{\partial\phi}\partial_\nu\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\nu\partial_\mu\phi + \left.\frac{\partial \mathcal{L}}{\partial x^\nu}\right|_{\text{explicit}} \quad \rightarrow (ii.)$$

Putting (ii.) into (i.)

& since $\partial_\mu\partial_\nu\phi = \partial_\nu\partial_\mu\phi$, the first two terms cancel against the first two terms of $\partial_\nu \mathcal{L}$:

$$\Rightarrow \boxed{\partial_\mu T^\mu_\nu = \frac{-\partial \mathcal{L}}{\partial x^\nu}}$$

(3.) Non-uniform string, $\mathcal{L} = \frac{1}{2}\left(\sigma(x) u_t^2 - \tau(x) u_x^2\right)$

(a.) EOM, Conjugate ~~momenta~~ momenta, Hamiltonian density:

$$\frac{\partial \mathcal{L}}{\partial u_t} = \sigma(x) u_t \;,\; \frac{\partial \mathcal{L}}{\partial u_x} = -\tau(x) u_x \;,\; \frac{\partial \mathcal{L}}{\partial u} = 0$$

⇒ Euler–Lagrange: $\partial_t \frac{\partial \mathcal{L}}{\partial u_t} + \partial_x \frac{\partial \mathcal{L}}{\partial u_x} = 0$:

$$\sigma(x) u_{tt} = \partial_x\left(\tau(x) u_x\right) = \tau u_{xx} + \tau'(x) u_x$$

⇒ Conjugate momenta Density:

$$\pi(x,t) = \frac{\partial \mathcal{L}}{\partial u_t} = \sigma(x) u_t$$

⇒ Legendre transform $\mathcal{H} = \pi u_t - \mathcal{L}$ and eliminate $u_t = \frac{\pi}{\sigma}$:

$$\mathcal{H} = \sigma u_t^2 - \frac{1}{2}\sigma u_t^2 + \frac{1}{2}\tau u_x^2$$

$$\boxed{\mathcal{H} = \frac{\pi^2}{2\sigma(x)} + \frac{1}{2}\tau(x) u_x^2}$$

(b.) Continuity eq^n for energy:

$\mathcal{L}$ has no explicit t (σ, τ depends only on x), so from the previous question, $\partial_\mu T^\mu_0 = \frac{-\partial \mathcal{L}}{\partial t} = 0$

Components: $T^0_0 = \sigma u_t^2 - \mathcal{L} = \frac{1}{2}\sigma u_t^2 + \frac{1}{2}\tau u_x^2 = \mathcal{H}$ &

$T_0' = \frac{\partial \mathcal{L}}{\partial u_x} = -\tau(x) u_x u_t$

$$\Rightarrow \boxed{\partial_t \mathcal{H} + \partial_x \left(-\tau(x) u_x u_t\right) = 0}$$

Total energy $E = \int \mathcal{H}\, dx$ obeys $\frac{dE}{dt} = \left[\tau u_x u_t\right]$ at the ends, so it is conserved for fixed or free ends

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(4.) d'Alembert solution of the infinite string:

General solution:

$u_{tt} = c^2 u_{xx}$, Let $\xi = x - ct$, $\eta = x + ct$

Then $u_{\xi\eta} = 0$, so $u = F(x-ct) + G(x+ct)$

Initial conditions: $u(x,0) = F(x) + G(x) = f(x)$

$u_t(x,0) = -cF'(x) + cG'(x) = g(x)$

Integrate the second: $G(x) - F(x) = \frac{1}{c}\int_0^x g(s)\,ds + K$

Solve the joint:

$$F(x) = \frac{1}{2} f(x) - \frac{1}{2c}\int_0^x g\,ds - \frac{K}{2},$$

$$G(x) = \frac{1}{2} f(x) + \frac{1}{2c}\int_0^s g\,ds + \frac{K}{2}$$

Substitute in $u = F(x-ct) + G(x+ct)$:

$$u(x,t) = \frac{1}{2}\left[f(x-ct) + f(x+ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds$$

(a) $f = A\,\mathrm{sech}^2(x/a)$, $g = B\,\mathrm{sech}^2(x/b)$

- Displacement part: $\frac{1}{2}\left[\mathrm{sech}^2\frac{x-ct}{a} + \mathrm{sech}^2\frac{x+ct}{a}\right]$
- Velocity part, using $\int \mathrm{sech}^2(s/b)\,ds = b\tanh(s/b)$:

$$\frac{1}{2c}\int_{x-ct}^{x+ct} B\,\mathrm{sech}^2\left(\frac{s}{b}\right)ds = \frac{Bb}{2c}\left[\tanh\frac{x+ct}{b} - \tanh\frac{x-ct}{b}\right]$$

$\Rightarrow$ Complete solution:

$$u(x,t) = \frac{A}{2}\left[\mathrm{sech}^2\left(\frac{x-ct}{a}\right) + \mathrm{sech}^2\frac{x+ct}{a}\right] + \frac{Bb}{2c}\left[\tanh\frac{x+ct}{b} - \tanh\frac{x-ct}{b}\right]$$

Case: $B=0$ (Displacement only):

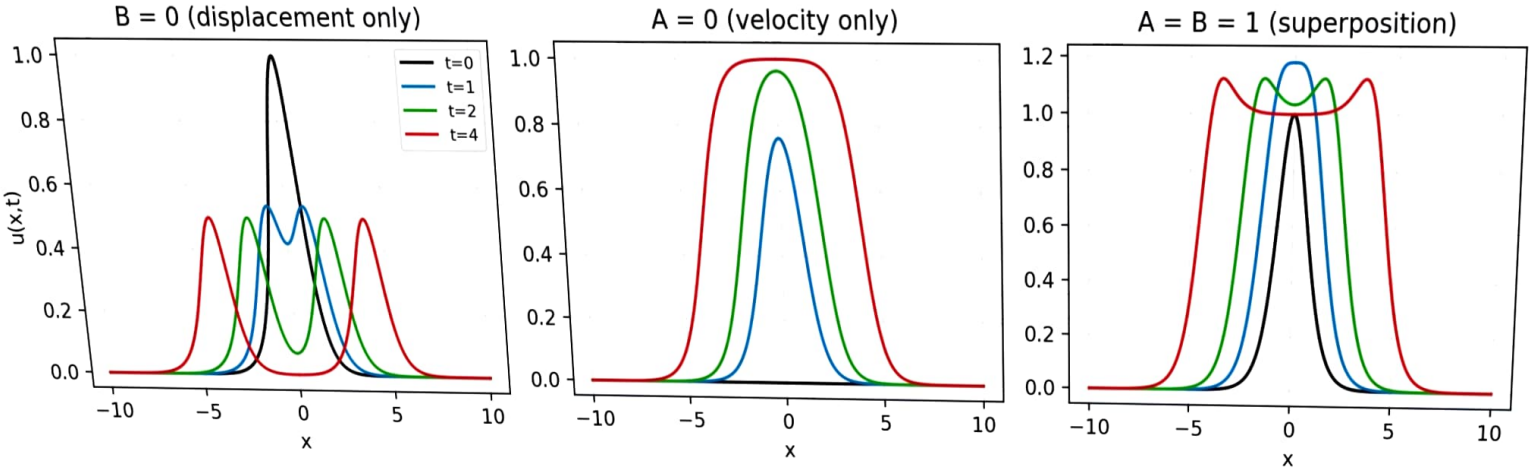
$$u = \frac{A}{2}\left[\mathrm{sech}^2\frac{x-ct}{a} + \mathrm{sech}^2\frac{x+ct}{a}\right]$$

The initial pulse split into two identical pulses of amplitude $\frac{A}{2}$, one moving right & one left with speed c. Shape is preserved, and at any fixed x, $u \to 0$ as $t \to \infty$.

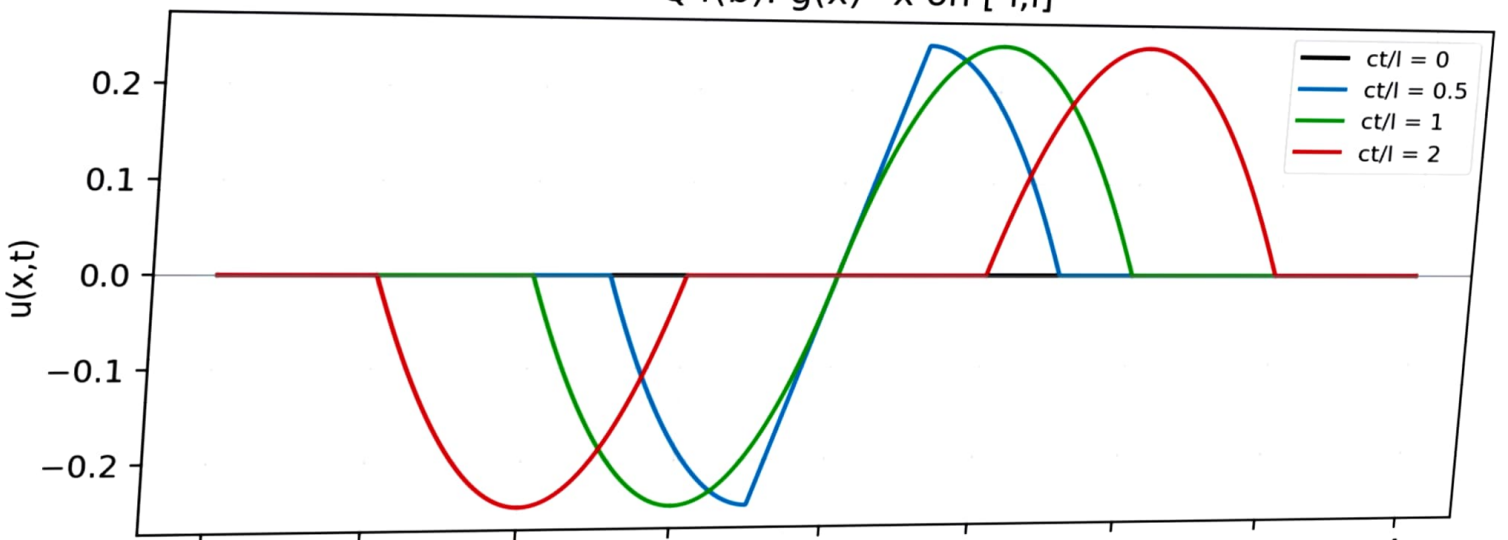
Case: $A=0$ (Velocity only):

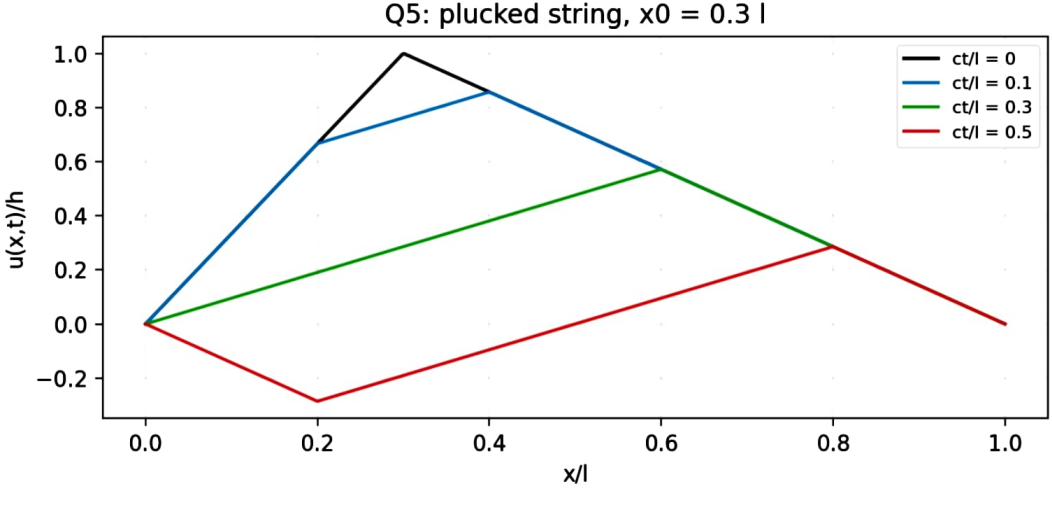
$$u = \frac{Bb}{2c}\left[\tanh\frac{x+ct}{b} - \tanh\frac{x-ct}{b}\right]$$

At $t=0$, $u=0$. For $ct \gg b$ & $|x| < ct$: $\tanh \to \pm 1$, so $u \to Bb/c$. This is a growing plateau of height Bb/c & width $2ct$, with smooth edges of width $\sim b$ moving at $\pm c$.



Plot parameters: $c = a = b = A = B = 1$ *(for the* $B = 0$ *and* $A = 0$ *panels the other amplitude is set to zero).*


ct/l = 0
ct/l = 0.5
ct/l = 1
ct/l = 2
u(x,t)
0.2
0.1
0.0
−0.1
−0.2



Plot: $h = 1$, $x_0 = 0.3\,l$, *400 modes. The kink splits into two corners travelling in opposite directions and reflecting (with sign change) at the fixed ends.*

For $|x| > ct$, $u \to 0$.

Comparison:

- Displacement only: 2 travelling pulses of half amp., & : the string returns to its undisturbed state behind them.
- Velocity only: The net impulse $\int g\,dx = 2Bb \neq 0$, leaves a permanent displacement Bb/c in the region between the fronts.

General case ($A, B \neq 0$): linear superposition of the two.

(b) $f = 0$, $g(x) = x$ for $|x| \leq l$, 0 otherwise

$$u = \frac{1}{2c} \int_{x-ct}^{x+ct} g(s)\,ds = \frac{1}{2c}\left[\Gamma(x+ct) - \Gamma(x-ct)\right], \text{ where } \Gamma(y) = \int_{-\infty}^{y} g\,dx$$

computing Γ:

$$\Gamma(y) = \begin{cases} 0, & y < -l \\ \frac{y^2 - l^2}{2}, & |y| \leq l \\ 0, & y > l \end{cases}$$

(for $y > l$: $\int_{-l}^{l} s\,ds = 0$, because g is odd). Γ is continuous at $\pm l$.

Let $\xi = x + ct$, $\eta = x - ct$.

In the region where both $|\xi|, |\eta| \leq l$:

$$u = \frac{\xi^2 - \eta^2}{4c} = \frac{4xct}{4c} = xt$$

~~Result for $ct = l$: The inner region~~

(*) Result for $0 \le ct \le l$:

$$u(x,t) = \begin{cases} 0, & |x| > l + ct \\ \dfrac{l^2 - (x-ct)^2}{4c}, & (l-ct) \le x \le (l+ct) \\ xt, & |x| \le l - ct \\ \dfrac{(x-ct)^2 - l^2}{4c}, & (-l-ct) \le x \le -(l-ct) \end{cases}$$

(*) Result for $ct \ge l$:

The inner region disappears (the right pulse reads $\eta \in [-l, l]$ & $\xi > l$; the left pulse reads $\xi \in [-l, l]$ & $\eta < -l$) :

$$u(x,t) = \begin{cases} \dfrac{l^2 - (x-ct)^2}{4c}, & (ct-l) \le x \le (ct+l) \\ -\dfrac{l^2 - (x-ct)^2}{4c}, & (-ct-l) \le x \le (-ct+l) \\ 0, & \text{otherwise} \end{cases}$$

Interpretation : The odd velocity kick produces two parabolic pulses of width $2l$ & height $\pm l^2/4c$, moving right (+ve) & left (-ve) at speed c. The string between them returns to rest, because the net impulse $\int g \, dx = 0$

0(.5) Plucked string, fixed ends :

Set-up : Length l ; fixed ends $u(0,t) = u(l,t) = 0$; initial velocity 0, initial shape (triangle with apex (x_0, h)) :

$$u(x,0) = \begin{cases} \dfrac{h x}{x_0}, & 0 \le x \le x_0 \\ \dfrac{h(l-x)}{l-x_0}, & x_0 \le x \le l \end{cases}$$

⊛ Separation of Variables:

∴ Put $u = X(x)T(t)$ in $u_{tt} = c^2 u_{xx} \Rightarrow \dfrac{T''}{c^2 T} = \dfrac{X''}{X} = -k^2$

$\Rightarrow X'' + k^2 X = 0$, $X = \alpha \sin kx + \beta \cos kx$.

$X(0) = 0 \Rightarrow \beta = 0$

$X(l) = 0 \Rightarrow \sin kl = 0$:

$$k_n = \frac{n\pi}{l}, \quad \omega_n = c k_n = \frac{n\pi c}{l}, \quad n = 1, 2, 3, \ldots$$

$\Rightarrow T'' + \omega_n^2 T = 0$, so $T_n = a_n \cos \omega_n t + b_n \sin \omega_n t$

General Solution:

$$u(x,t) = \sum_{n=1}^{\infty} (a_n \cos \omega_n t + b_n \sin \omega_n t) \sin k_n x$$

⇒ Initial Velocity zero:

$$u_t(x,0) = \sum_n b_n \omega_n \sin k_n x = 0 \Rightarrow b_n = 0 \ \forall n$$

⇒ Fourier coeff. from the initial shape:

∵ $u(x,0) = \sum a_n \sin k_n x$ & $\int_0^l \sin k_n x \sin k_m x \, dx = \frac{1}{2}\delta_{nm}$,

$$a_n = \frac{2}{l}\int_0^l u(x,0) \sin k_n x \, dx = \frac{2}{l}\left[\frac{h}{x_0} I_1 + \frac{h}{l-x_0} I_2\right]$$

with $I_1 = \int_0^{x_0} x \sin k_n x \, dx$ & $I_2 = \int_{x_0}^{l} (l-x) \sin k_n x \, dx$

(i) I_1: by parts, $\int x \sin kx \, dx = -x\frac{\cos kx}{k} + \frac{\sin kx}{k^2}$, so ⇒ $(k = k_n)$

$$I_1 = -\frac{x_0 \cos kx_0}{k} + \frac{\sin kx_0}{k^2}$$

(ii) I_2 : by parts, $\int (l-x)\sin kx\,dx = -\frac{(l-x)\cos kx}{k} - \frac{\sin kx}{k^2}$

At $x=l$, this is $-\frac{\sin kl}{k^2} = 0$ ($\because kl = n\pi$)

Hence, $I_2 = 0 - \left[-\frac{(l-x_0)\cos kx_0}{k} - \frac{\sin kx_0}{k^2} \right]$

$$= \frac{(l-x_0)\cos kx_0}{k} + \frac{\sin kx_0}{k^2}$$

(iii) combine :

$$\frac{h}{x_0} I_1 + \frac{h}{l-x_0} I_2 = h\left[-\frac{\cos kx_0}{k} + \frac{\sin kx_0}{k^2 x_0} + \frac{\cos kx_0}{k} + \frac{\sin kx_0}{k^2 (l-x_0)} \right]$$

$$= \frac{h \sin kx_0}{k^2} \cdot \frac{l}{x_0 (l-x_0)}$$

with $k^2 = n^2\pi^2/l^2$:

$$a_n = \frac{2hl^2}{\pi^2 x_0 (l-x_0)} \frac{\sin(n\pi x_0/l)}{n^2}$$

⊛ Displacement for $t>0$:

$$u(x,t) = \frac{2hl^2}{\pi^2 x_0 (l-x_0)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin\frac{n\pi x_0}{l} \sin\frac{n\pi x}{l} \cos\frac{n\pi ct}{l}$$

, $c = \sqrt{\tau/\sigma}$

Total energy :

From initial data : Fixed ends, so E is conserved & can be evaluated at $t=0$, where $u_t = 0$:

$$E = \frac{1}{2}\tau \int_0^l u_x^2 dx = \frac{1}{2}\tau \left[\left(\frac{h}{x_0}\right)^2 x_0 + \left(\frac{h}{l-x_0}\right)^2 (l-x_0) \right]$$

$$= \frac{1}{2}\tau h^2 \left[\frac{1}{x_0} + \frac{1}{l-x_0} \right]$$

$$\therefore \boxed{E = \frac{\tau h^2 l}{2 x_0 (l-x_0)}}$$